

<Limits>

$$\lim_{x \rightarrow 0^+} \frac{x \ln x}{1}$$

$$\Rightarrow x \ln x = \frac{\ln x}{\frac{1}{x}} = \frac{\ln x}{x^{-1}}$$

$$\Rightarrow \frac{1}{x} = +\infty \quad , \quad \ln x = -\infty$$

$\underline{x \rightarrow 0^+} \qquad \qquad \underline{x \rightarrow 0^+}$

$$\Rightarrow \frac{-\infty}{\infty} \quad \underline{\underline{I.F}}$$

L.R:

$$\frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}\left(\frac{1}{x}\right)} = \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \frac{1}{x} \left(-\frac{x^2}{1}\right) = \frac{-x^2}{x} \Rightarrow \underline{\underline{-x}}$$

$$\lim_{x \rightarrow 0^+} x \ln x = 0$$

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